An audit of Astra's notes, and one combined form of the proof

A fourth research note on "The maximum number of cycles in a graph of large rank"

Fable 5.2, for KNT — Research note of 12 November 2026

KNT gave me notes of Astra with eight proposals for a shorter proof, and asked which of them can be incorporated. This note has the audit, and it describes one copy of the paper that contains the sound parts together with the reductions of my three notes [N1], [N2], [N3]. [M] is the draft of 42 pages; Astra's notes use its numbers. [T] is the new copy, `trial-combined.pdf`. The manuscript is not changed.

The result in eight sentences

1. *The audit.* I checked every step of the eight proposals, and I found no error. One step needs an exact reading, which Section 2.3 gives.
2. *Agreement.* Three proposals are the same as reductions of my notes: the marked set, the gap theorem as a corollary, and the recursion from a length d to a length of order $\log d$. Astra's notes do not mention [N1], [N2], [N3], so I take this as two independent derivations.
3. *New and in the copy.* The lemma on a shortest path, which needs no hypothesis on the number of paths; the proof of rigidity by a limit; the cut of size three that excludes three full pairs at a port; the upper bound for pairs with two links; a shorter proof of nesting; a shorter window step in the lemma on layers.
4. *Sound, but not in the copy.* Caps for cuts of size two and three: measured, 11 lines more. Connected blocks in the chain: measured, 1 line more. The form of the theorem on cuts of size four with two sides: not necessary when one side is a cycle of length four. The recursion with two deleted sets: the form of [N1] has one family of cycles.
5. *Measured.* The copy [T] has 41 pages, 1577 typeset lines, 39 numbered statements and three cases. The manuscript has 42, 1624, 40 and five. The copy of [N3] has 41, 1587, 39 and three.
6. *A price of [N3] is gone.* With the limit proof, the gap constant is used at one place. Proposition 6.3 keeps the proof and the numbers of the manuscript, and the lemma on a full star is not necessary.
7. *Astra's notes keep the case of bounded girth.* With [N2] the statements that the notes list as retained ([M, Lemma 5.5], [M, Corollary 5.6], [M, Section 7]) are not necessary. The copy [T] uses [N2].
8. *Status.* All new proofs are on paper. Nothing of this note or of Astra's notes has a formal proof, and the text of [T] differs from the Lean proof in more places than [M].

NOTATION | the status of this note

- *Checked on paper.* Each proposal, in the form of Astra's notes. The copy [T] has the proofs of the adopted parts in full, and I read the typeset text of its Sections 4 to 7 again.
- *Tested by computer.* The script `checks/astra_audit_check.py` (Section 5). The tests cover small graphs and the arithmetic. They do not replace a proof.
- *Measured.* Every number of lines in this note is from a built copy.
- *Not done.* No formal proof. No reader other than Astra, for its own proposals, and me. I did not build a copy without [N2] (Section 6).

Contents

1 Verdicts . . . 2
2 The audit, proposal by proposal . . . 3
3 The combined copy . . . 6
4 The text and the formal proof . . . 7
5 Tests . . . 7
6 Open points . . . 7
7 Files . . . 8

1 Verdicts

	Proposal	Sound	In [T]	Measured effect, remark
1	Marked set; no edge recursion; five marked vertices at the end	yes	yes	same as reduction 4 of [N3]; -2 lines
2	Window step with the threshold $m^{3/4}$	yes	yes	Section 4: -5 lines
3	A shortest path: counts without a hypothesis; cycles by their edges on the path	yes	yes	Section 5: -1 line; no ordinary layers, no matchings, one conclusion in [M, Proposition 5.3]
4	A gap with some fixed constant, without induction	yes	yes	same as reduction 1 of [N3]; -24 lines there
5	Caps for cuts of size two and three; the bound $2^{r-1} + 420 \cdot 2^{r/2}$	yes	no (option)	$+11$ lines
6	Rigidity by limits; three full pairs give a cut of size three	yes	yes, for a cycle of length four	Section 6: -6 lines; no lemma on a full star
7.1	Shorter proof of nesting	yes	yes	with 7.2: -3 lines
7.1	Connected blocks	yes	no (option)	$+1$ line
7.2	Two links as an upper bound for W_2	yes	yes	shorter statement; shorter proof for heavy families
7.3–7.5	Packing at logarithmic girth; recursion $d \mapsto C_0 \log(Qd + 2)$; direct end	yes	no	same recursion as [N1]; [N1] has one family of cycles, and [N2] needs another base
8.1	Coarse bound for the dyadic scales	yes	no	no dyadic scales in [N1]
8.2	Fixed blocks in [M, Proposition 5.3]	yes	no	Astra does not recommend it

The effects are differences of typeset lines against the copy of [N3] (lean form of [N2] with the four reductions), section by section. The introduction of [T] is five lines longer, because it states the origin of the new parts.

2 The audit, proposal by proposal

2.1 Proposal 1: the marked set

For a connected cubic multigraph without a bridge and a set P of k vertices, Astra proves $\beta(G-P) \le r - \lceil k/2 \rceil - 1$ and $c(G) \le \big(q(G,P) + 2^{-\lceil k/2 \rceil - 1}\big)2^r$. This is the lemma on cycles and a marked set of [N3], with the integer part. The proof is the same count. I checked the two uses: $|P| = 2\lceil \log r \rceil$ gives the term $\frac{1}{2r}$ for the leading term, and five vertices give $\frac{2}{5} + \frac{1}{16} < \frac{1}{2}$ in the last case. The copy [T] has eight vertices there, as [M]; both numbers are correct.

2.2 Proposal 2: the window step

The step uses three inequalities, and all three hold: $(t + \frac{1}{6})^4 \le t^4 + t^3$ for $t \ge 1$, because $\frac{2}{3} + \frac{1}{6} + \frac{1}{54} + \frac{1}{1296} < 1$; $6\mu^{3/4} + 3k(6\mu^{1/4} + 3) \le 33\mu^{3/4}$ for $k \le \sqrt{\mu}$; and $132 \log^2 \mu \le \mu^{1/4}$ for $\mu \ge 2^{100}$, because $132 \cdot 10^4 < 2^{25}$ and the right side grows faster. Step 1 of [M, Lemma 4.1] and the number $\Theta_\delta(k)$ do not change. This is [T, Lemma 4.1, Step 2].

2.3 Proposal 3: a shortest path

Let N be a bridgeless 2-pole of rank $b \ge 2$, $A = w_0 \cdots w_\ell$ a shortest path between its ports, $J = N - V(A)$ and $\Delta = b - \ell$. Astra's statements, in the form of [T, Lemma 5.3]:

(a) $0 \le \Delta \le \beta(J)$;

(b) at most 2Δ layers between the ports are not thin, and at most 4Δ vertices of J lie outside the thin layers;

(c) every cycle of J has at most $4\Delta + 2$ vertices;

(d) $c(N) \le 10 \cdot 2^{5\beta(J) + b/2}$.

I checked (a), (b), (c) as written. In (d), the restriction of an even subgraph to $E(A)$ has the kernel $\mathcal{Z}(J)$, so at most $2^{\beta(J)}$ cycles have the same edges on A; this is correct also when an edge of A is doubled.

> **IDEA | the step that needs an exact reading**
>
> Astra's notes say: if the layers at both ends of an edge e_i of A are thin and e_i is absent from the cycle, then the further edge of w_{i-1} points forward and that of w_i points backward, and "positions satisfying this condition cannot be consecutive". The condition must be read as a property of the *graph*: call i free if the two further edges have these directions. Then no two consecutive indices are free, an index that is not free between thin layers has its edge in every cycle that passes it, and the count $2^{\lceil (z-a-1)/2 \rceil}$ is correct. Read as a property of the cycle ("no two consecutive absent edges") the count would be false: there are more than $2^{n/2}$ such patterns. The copy [T] defines free indices.

With (d), [M, Theorem 5.4] has a proof of five lines and the constant $K_\varepsilon = 20 \cdot 2^{5C_\varepsilon}$. [M, Proposition 5.3] keeps one conclusion, $\beta(J) \le C_\varepsilon$, and Steps 1 to 3 of its proof. The ordinary layers, the count of perfect matchings and the items (ii), (iii) are not necessary. Astra's notes keep the bound for the layers that are not thin, because [M, Corollary 5.6] uses it. With [N2] that corollary is not in the text, so (b) is used only inside the lemma.

In lines this proposal is neutral: Section 5 has 275 lines in [T] and 276 in the copy of [N3]. I put it in the copy for its content. The counts that hold for every pole are now in one lemma, and the proposition with the forest estimate gives one number.

2.4 Proposal 4: the gap theorem

Astra's proof is that of [N3]: a bridgeless 2-pole is a capped ladder if and only if J is a forest, and a cycle of J gives a gap by [M, Lemma 5.1]. The two notes bound the length of the cycle in different ways. [N3] takes it from Step 2 of [M, Proposition 5.3]. Astra takes it from (c) and $\Delta \le \beta(J) \le C_{1/4}$, which gives $\gamma_0 = 2^{-4C_{1/4}-2}$. The copy [T] has Astra's form ([T, Theorem 5.7]), because (c) is there in any case. Astra's warning is correct: the old arithmetic with 3.99 cannot use the new constant without change. [N3] changed the arithmetic. The copy [T] does not need it (Section 2.6).

2.5 Proposal 5: caps for cuts of size two and three

The proposal is correct. I checked the cap for two and for three ports, the inequalities $z < 4\eta$ and $\mathcal{B} > 2^{d-2} - 3\eta$, the step from $a_1 + a_2 \ge \frac{2}{3}\sum_j a_j$ to $a_3 = o(1)$, the bounds $c(N) \le 20 \cdot 2^{b/2}$ for a pole with a full pair and $P_{ab} \le 20\sqrt{2} \cdot 2^{h/2}$ for a 3-pole with two full pairs, and the constant: the excess is at most $(15 + 400)\, 2^{r/2}$.

I built it as an option of the script (`caps`). The copy has 1588 lines, 11 more than [T]. The reason is that the proof of [M, Proposition 6.3] does not need full pairs. It uses [M, Theorem 5.4] and one gadget, and it does not use the gap theorem or a cap. The proposal has its merits: one method for all cuts, an absolute constant 420, and no numbers $\frac{1}{1000}$, K_0, $\kappa(h)$ in the paper. KNT can choose it; the proofs are in the option.

2.6 Proposal 6: rigidity by limits

The three implications. For the limit of the states of a sequence of 4-poles whose ranks tend to infinity, Astra states

$$a_{ik} + a_{jk} > 1 \implies a_{ij} = 0, \qquad a_{kl} > 0 \implies a_{ij} + F - u_{ij|kl} \le 1,$$
$$a_{ik} + a_{il} + a_{jk} + a_{jl} > 2 \implies u_{ik|jl} + u_{il|jk} = 0.$$

They need the gadgets of [M, Lemma 5.7] for every positive margin, as Astra says. So [T, Lemma 5.9] is stated for every ε; its proof is that of [M].

The classification. I checked the proof that a limit with $S + F \ge 4$ is a cross or a star, line by line, and a script enumerated all states on three grids (Test 3). There is no other state.

Three full pairs. Astra excludes them without [M, Lemma 6.7]: the pole without the two other ports is a capped ladder, the two other ports cannot lie in its first end piece, and this piece then has three edges to the rest of the graph. I checked the proof, and Test 2 examined all positions of two ports in the capped ladders of weight 3 to 6.

What [T] takes. The copy has the theorem for a cycle C of length four (reduction 2 of [N3]). There the count is $c(G)\,2^{-k} = 2^{-k} + z + (S + F - u_{12|34}) + (S + F - u_{14|23})$, and one use of the cap gives $S + F \geq 4 - \rho_0$. So [T] has: the cap with a free error η ([T, Lemma 6.4]); the rigidity lemma with the statement of [M] and Astra's proof by a limit ([T, Lemma 6.5]); and Astra's cut of size three as Case 1 of [T, Theorem 6.6]. The second cap, the bound for z, the lemma on a full star and the arithmetic of the star case are not necessary. Astra's step 6.5 (the full pairs of the smaller side) is not necessary either, because the other side is C.

The constants. In [N3] the gap constant forced $\varepsilon_0 = \gamma_0/20$ in all of Section 6. In [T] the gap theorem is used in the last step of the limit proof only. So [M, Proposition 6.3] keeps its proof and its numbers 0.501 and 1.002.

2.7 Proposal 7: the last section

7.1, nesting. The proof with the two cases $D \not\subseteq U_C$ and $D \subseteq U_C$ is correct, and it is in [T]. *Connected blocks:* for $j \geq i + 2$ the separators between C_i and C_j lie in one component of $H - (C_i \cup C_j)$; I checked the argument with a path from u to p. This removes the choice of a large component, and the factor 2Λ in the thresholds. The proof of the one component has the same length as the choice of a component: the option `blocks` gives 1578 lines.

7.2, two links. The bound $W_2 \leq 2\mu(e + NM)$ is correct, with M a bound for the weight of the regions that separate a neighbour u of A from p. It is the lemma of [M] in the other direction, with no condition on μ and no number K. It is [T, Lemma 7.9], and the proof for heavy families is shorter with it. Test 4 compares the two sides for random families.

7.3 to 7.5. I checked the packing bound $C_2 d^6 (s + 1) 2^d$ with $m = 1$, the recursion $|S_d| + 1 \leq 2^{2d}$ with $a(d) = \lceil C_0 \log(Qd + 2) \rceil$, and the end with $\mu \geq \frac{5}{3}$ and $\mathbb{P}(Z > 0) \geq \frac{5}{8} - o(1)$. They are correct. The recursion from d to a length of order $\log d$ is the idea of [N1]. The two forms differ in one point: Astra deletes two nested sets $S^- \subseteq S^+$, and [N1] deletes one set and counts the blocks that contain a short cycle. The direct end is the inequality of the proposition on heavy families. I keep the form of [N1], because [N2] is written for it: in [N2] the start of the recursion is girth five, not a large girth D_*.

2.8 Proposal 8 and the list of retained parts

Proposal 8.1 is correct and has no use when there are no dyadic scales. Proposal 8.2 is correct; Astra does not recommend it, and I did not build it.

Astra's notes keep [M, Lemma 5.5], [M, Corollary 5.6] and [M, Section 7], and they say that no replacement was found. [N2] is such a replacement: a large piece of girth five has few forests, so [M, Theorem 8.1] holds for girth five and the case of bounded girth is empty. Astra's notes do not refer to [N2], so I take it that Astra has not audited it.

3 The combined copy

Section of [T]	Content	Origin of the changes
3 Forest estimate	as [M]; Lemma 3.7 on a marked set; the leading term from it	[N3], Astra 1
4 Layers	Lemma 4.1 with the new Step 2	Astra 2
5 Poles	Lemma 5.3 (a shortest path); Proposition 5.4 ($\beta(J) \leq C$); Theorem 5.5 (few cycles); Lemma 5.6 (ladders, three equivalent statements); Theorem 5.7 (gap); Corollary 5.8; Lemma 5.9 (three gadgets, every margin)	Astra 3, 4; [N3]
6 Rings and cuts	rings as [M]; Proposition 6.3 as [M], also for loops; Lemma 6.4 (cap); Lemma 6.5 (rigidity, by a limit); Theorem 6.6 (a cycle of length four)	[N3]; Astra 6
7 Girth five	lean form of [N2]; Lemma 7.7 (nesting) with the short proof; Lemma 7.9 (two links) as an upper bound	[N1], [N2]; Astra 7.1, 7.2
8 Proof	three cases	[N3]

Against [M], the copy does not have: Section 7 (bounded girth), [M, Lemma 5.5], [M, Corollary 5.6], [M, Lemma 5.9] (a third port), [M, Lemma 6.4] (gluing), [M, Lemma 6.7] (full star), [M, Theorem 8.8] (raising the girth), the ordinary layers, and the edge recursion. It has these new statements: the lemma on a marked set, the lemma on a shortest path, the two lemmas of [N2] on full pairs and rigid sets at girth five with the lemma on pieces, and the proposition on heavy families.

3.1 Measured

	[M]	[N3] copy	[T]	[T], caps	[T], blocks
Pages	42	41	41	41	41
Typeset lines	1624	1587	1577	1588	1578
Numbered statements	40	39	39	39	39
Cases of the main theorem	5	3	3	3	3
Section 4 (layers), lines	72	73	68	68	68
Section 5 (poles), lines	339	276	275	278	275
Section 6 (rings and cuts), lines	275	258	252	260	252
Bounded girth and last section, lines	323	358	355	353	356
Proof of the main theorem, lines	30	20	20	20	20

The numbers of statements are equal because one lemma comes (a shortest path) and one goes (a full star). The gain of [T] against the copy of [N3] is ten lines. Its larger gain is in the content: no numbers that depend on the gap constant, no lemma on a full star, and the counts in a pole separated from the forest estimate.

4 The text and the formal proof

Nothing of Astra's notes or of my notes is formalised. In [M] the text and the Lean proof differ in four places. In [T] they differ in these places also; Section 1.4 of [T] says so.

- The leading term: the formal proof has the edge recursion.
- [T, Theorem 5.5]: the formal proof counts perfect matchings between ordinary layers.
- [T, Theorem 5.7]: the formal proof is an induction with a third port, and it gives $\frac{1}{16}$.
- [T, Lemma 6.5] and [T, Theorem 6.6]: the formal proof treats a cyclic cut of size four with two sides; it has the arithmetic with fixed numbers and the bound for a full star.
- The last case: the formal proof has the case of bounded girth and two girth scales.

The formal statement of the main theorem does not depend on any of this.

5 Tests

The script `checks/astra_audit_check.py` has six tests. Exit code 0 means that all passed. With the arguments `3000 2` it prints:

Test	Content and result
1	[T, Lemma 5.3] for 3000 random bridgeless 2-poles of rank 2 to 7 and all 4157 shortest paths: (a), (b), (c); at most $2^{\beta(J)}$ cycles with the same edges on A; no two consecutive free indices; the edge of an index that is not free between thin layers is in the cycle; the bound of (d). The largest value of $c(N)\,2^{-\beta(J)-4\Delta-\ell/2}$ was 4.125; the lemma allows 10.
2	9354 capped ladders of weight 3 to 6 with two more ports on their edges. In 282 of them both pairs at the cap are full. In each of these, no port is in the first end piece, and two edges leave it.
3	All states with entries in the grids of step $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{8}$ that satisfy the three implications, $F \le 1$ and $S + F \ge 4$: 361 states, 9 crosses and 352 stars.
4	$W_2 \le 2\mu(e + NM)$ for 300 random families of regions with weights in cubic graphs (5752 ordered pairs with two links). The largest ratio of the two sides was 0.263.
5	The two inequalities of the window step, for $t \in [1, 101]$ and $\log \mu \in [100, 20000]$.
6	The caps of poles with two and three ports: the count of the cycles through the cap, 150 poles.

Tests 1, 2, 4, 6 are complete counts on small graphs. Test 3 is a finite part of a statement about real numbers. The rigidity lemma and the theorem on a cycle of length four contain large constants, and no computation tests them.

6 Open points

- *The base.* The copy [T] uses [N2] (no case of bounded girth) and the cycle of length four of [N3], which needs [N2]. Astra's notes are written for [M] with its Section 7. The proposals 2, 3, 4 and 7.2 do not depend on this choice. Proposal 6 then needs its form with two sides (Astra's steps 6.3 to 6.5, which I checked on paper), and [M, Corollary 5.6] needs (b) of the lemma on a shortest path. I did not build this copy.

- *Readers.* [N2] has had one reader. Its two lemmas at girth five carry the removal of a whole case, and they have very large constants. A second reader for them has more value than for any other part.
- *The formal proof.* A formal proof of [T] is new work in Sections 5 and 6 and in the last section. The existing Lean proof stays valid for the theorem.
- *The options.* `caps` and `blocks` are complete texts. Each is a choice of KNT; I recommend neither, by the measured lines.

7 Files

File	Content
`research/astra-audit.tex, .pdf`	this note
`research/apply_combined.py`	writes the combined form into a copy; options `caps` and `blocks`; every replacement must match exactly once
`research/trial-combined.pdf`	the copy [T], 41 pages
`checks/astra_audit_check.py`	the six tests of Section 5

To build [T], copy the folders `exact-maximum` and `style` to a new folder. On the copy of `exact-maximum` run the three scripts in this order:

```
python3 apply_girth_five.py <copy> lean
python3 apply_fewer_cases.py <copy> keepgap
python3 apply_combined.py <copy>
```

Then build `exact-maximum.tex` with `tectonic`.

References

[M] KNT, Astra and Fable 5.2, *The maximum number of cycles in a graph of large rank*, draft of 12 November 2026, 42 pages, `exact-maximum/exact-maximum.pdf`.

[T] The combined copy of this note, `research/trial-combined.pdf`.

[N1] Fable 5.2, *The large-girth theorem with one family of short cycles*, `research/one-scale.pdf`.

[N2] Fable 5.2, *Pieces at girth five: the case of bounded girth is not necessary*, `research/girth-five.pdf`.

[N3] Fable 5.2, *Three cases: four reductions in the cut analysis and the final count*, `research/fewer-cases.pdf`.